

Comparison of Sigma-Delta-Over-Fiber and Analog Radio-Over-Fiber mm-wave Links With Optical Up-Conversion

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Abstract *Radio-over-fiber with central frequency up-conversion is promising for phase-coherent D-MIMO transmission, but extension to mm-wave requires impractically high sampling rates. We employ optical up-conversion to relax this constraint and compare ARoF and SDoF, showing SDoF performs better below 300 MBd. ©2026 The Author(s)*

Introduction

Radio-over-fiber (RoF) is a technology in which radio-frequency (RF)-signals are modulated onto an optical carrier, enabling the use of optical fiber cables to distribute the radio signals across long distances. It has been shown in^{[1]–[3]} that RoF is a particularly competitive solution to connect multiple distributed remote radio heads (RRHs) to a central-unit (CU) in a distributed MIMO (D-MIMO) network architecture. The optical fiber then carries modulated RF-signals between multiple RRHs and a common CU, where coordination and joint processing take place. The highest capacity gain from D-MIMO is achieved if the output signals from the separate RRHs are RF-phase coherent^[4].

Different methods can be used to encode the radio signal onto the optical carrier, where sigma-delta-over-fiber (SDoF) and analog RoF (ARoF) provide large benefits in D-MIMO networks, due to their inherently simple RRH design^[5]. Both ARoF and SDoF can be implemented with central frequency up-conversion to ensure RF-phase coherent transmission from the RRHs^[3]. Central frequency up-conversion means that the signals are up-converted to RF in the CU. This leads to a higher required CU sampling rate compared with conventional radio architectures, which typically transmit low-frequency baseband signals over the fiber. For SDoF the CU sample rate also should provide a satisfactory oversampling ratio for the sigma-delta modulation^[6].

To meet the capacity demands of future wireless systems, there is a shift toward higher carrier frequencies, as the available bandwidth increases

with frequency. Nevertheless, as the carrier frequency increases, implementing central frequency up-conversion becomes more challenging, since the required CU sampling rate grows impractically large. An alternative approach that reduces the sample rate requirement is to use a lower intermediate frequency (IF) for the radio signal and mix it with a sinusoidal signal in a photodetector to produce the desired millimeter wave (mm-wave) carrier, so called optical heterodyning^[7]. The sinusoidal signal is also modulated onto the optical carrier, but with certain frequency separation from the IF signal, such that the photodetector output signal ends up at the desired mm-wave carrier. Multiple demonstrations of RoF links using optical heterodyning to generate mm-wave carriers have been done for both ARoF^{[7]–[9]} and SDoF links^{[10]–[12]}. However, neither of these demonstrations has compared SDoF and ARoF mm-wave links with optical up-conversion.

In this paper, we compare the performance of SDoF and ARoF links, in which we use optical heterodyning to produce a 28 GHz carrier for wireless transmission. We show that, at low symbol rate, SDoF gives better performance in terms of measured error vector magnitude (EVM) due to its higher resilience to modulator non-linearities. At high symbol rate, we measure lower EVM with ARoF because SDoF is limited by quantization noise.

Radio-Over-Fiber Link

In Fig. 1, we present the RoF link used in the measurements. The same hardware is used for ARoF and SDoF, with the only difference being the out-

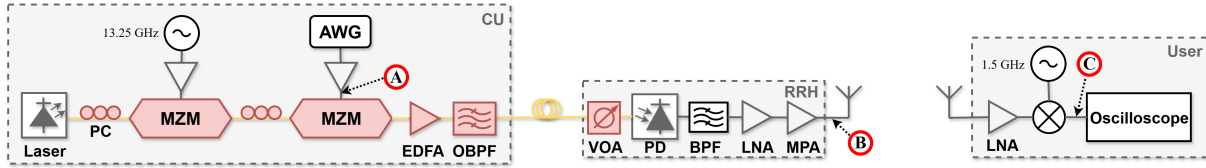


Fig. 1: The radio-over-fiber link with optical up-conversion. The central unit (CU) includes: laser, polarization controller (PC), Mach-Zehnder modulator (MZM), arbitrary waveform generator (AWG), erbium-doped fiber amplifier (EDFA) and optical band-pass filter (OBPF). The remote radio head (RRH) includes: variable optical attenuator (VOA), photodetector (PD), band-pass filter (BPF), low noise amplifier (LNA) and medium power amplifier (MPA)

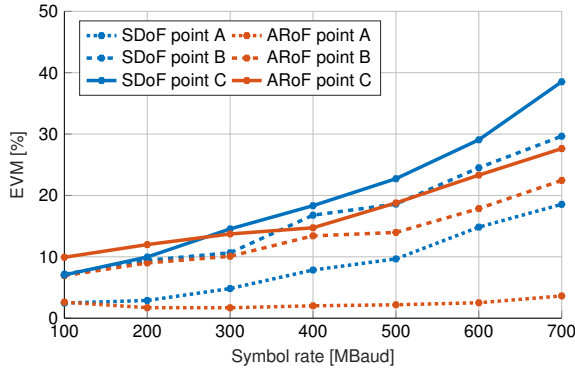


Fig. 2: Measured error vector magnitude (EVM) versus symbol rate for three different measurement points the analog radio-over-fiber (ARoF) and sigma-delta-over-fiber (SDoF) links specified in Fig. 1: electrical B2B 1.5 GHz (point A), optical 28 GHz (point B) and over-the-air (OTA) (point C).

put signal from the arbitrary waveform generator (AWG). In the ARoF measurements, the AWG generates a single-carrier signal at a carrier frequency of 1.5 GHz, while in the SDoF measurements, it generates the sigma-delta modulated version of the same signal. Two level sigma-delta modulation of 4th order is performed in Matlab, with a sample rate of 12 GS/s^[6]. See^[12] for further details of the signal generation.

The 1550 nm laser output is polarization-controlled and fed into a Mach-Zehnder modulator biased at the minimum transmission point. An RF signal of 13.25 GHz frequency is inserted to the electrical input of the modulator, to generate two sidebands separated by 26.5 GHz at the output. The average electrical input power to the first modulator is 13 dBm, which gives the highest carrier suppression. The output signal from the modulator is inserted to another polarization controller and Mach-Zehnder modulator biased at the quadrature point for maximum linearity. At the electrical input of the second modulator the information signal is inserted. The average electrical input power to the second modulator is set to the value that gives the lowest EVM at the output of the medium power amplifier (MPA) (point B in Fig. 1), which is 12 dBm for ARoF and 13 dB for SDoF, due to its higher resilience to modulator non-linearities^[5]. The output signal from the

second modulator is amplified with an erbium-doped fiber amplifier, filtered with an optical band-pass filter, and transmitted to the RRH via optical fiber cable. In the RRH, the signal power is controlled and attenuated prior to the photodetector to prevent saturation. The electrical output signal from the photodetector is inserted into an electrical band-pass filter and amplified with a low noise amplifier (LNA) and an MPA, before being inserted into the antenna port. At the user, the signals at a 28 GHz carrier are amplified with an LNA, down-converted to 1.5 GHz using an analog mixer and captured with an oscilloscope.

Experimental Results

We measure the EVM at three different locations in the links: at the output of the amplifier following the AWG for the 1.5 GHz IF signal (point A), for the 28 GHz signal at the output of the MPA (point B) and after wireless transmission of the 28 GHz signal (point C). We present in Fig. 2 the EVM as a function of symbol rate, comparing SDoF and ARoF at measurement point A–C. First, we observe that the EVM increases with the symbol rate, as a consequence of the degrading signal-to-noise ratio (SNR), as we use a fixed average signal power. In addition, for SDoF, also the in-band quantization noise increases with symbol rate due to decreasing oversampling-ratio causing the EVM to increase at a higher rate. In summary, at point C the lowest EVM is achieved by SDoF below 300 MBd and by ARoF above 300 MBd.

In Fig. 3, we present the frequency spectrum of the output and error signal for ARoF and SDoF in point B at 100 Mbaud and 400 Mbaud. The error spectrum is the frequency domain representation of the difference between the transmitted and received constellation symbols, hence it captures noise and distortion. For the 100 Mbaud signal both SDoF and ARoF are limited by thermal system noise, as the error spectrum is flat and in the same level as the out-of-band noise floor. For the 400 MBd signal we observe that the SDoF error spectrum has the typical wing-shape

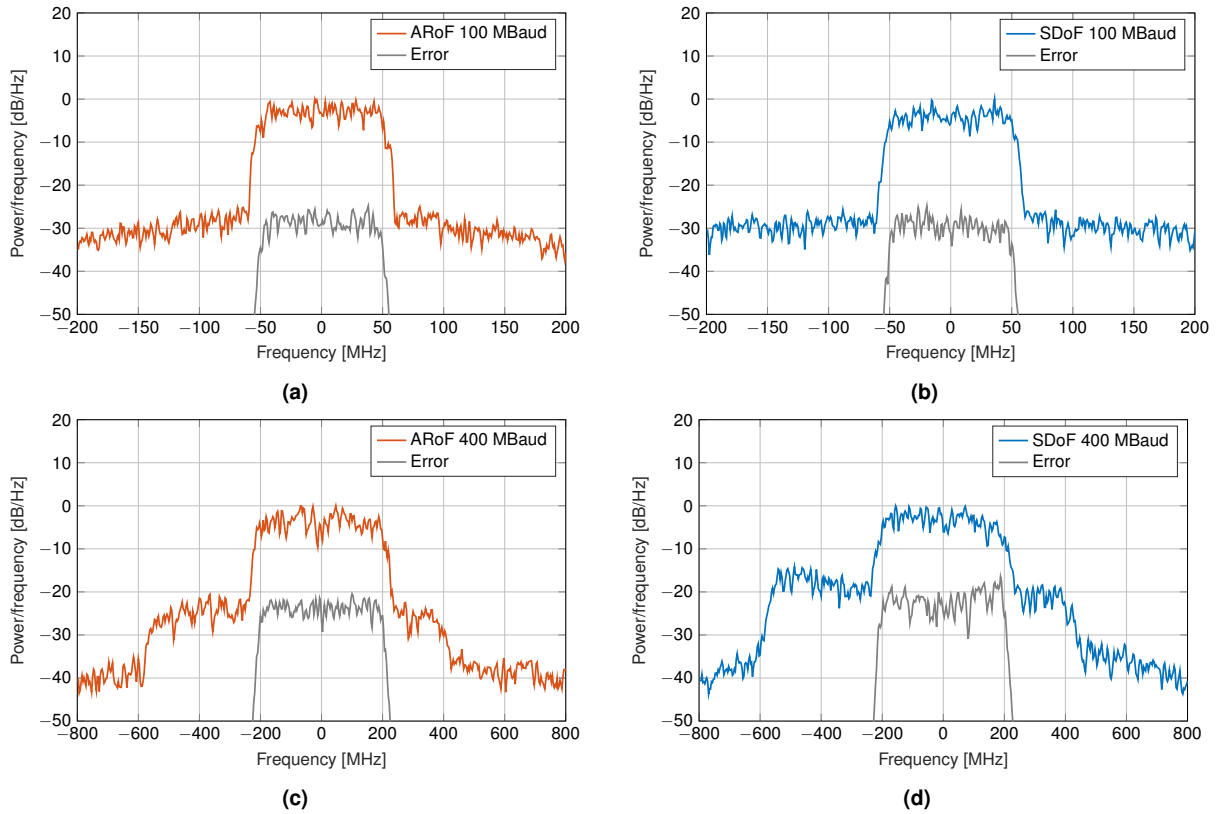


Fig. 3: Frequency spectrum of the output signal from the medium power amplifier (point B in Fig. 1), including the error signal. Two different symbol rates are shown; 100 MBd (a) analog radio-over-fiber (ARoF) and (b) sigma-delta-over-fiber (SDoF) and 400 MBd (c) ARoF and (d) SDoF.

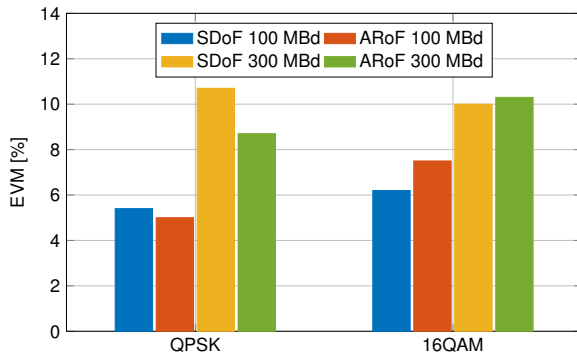


Fig. 4: Measured error vector magnitude (EVM) in over-the-air measurements (point C in Fig. 1) with two different orders of QAM modulation (QPSK and 16QAM) and two different symbol rates (100 and 300 MBd).

of sigma-delta quantization noise, while for ARoF it remains flat. Thus, at 400 MBd, the quantization noise significantly limits SDoF.

The reason that we measure a lower EVM with SDoF than ARoF below 300 MBd is that SDoF is less sensitive towards non-linearities in the modulator. As a consequence we can insert higher electrical power to the second modulator, resulting in a higher SNR at the receiver^[5]. This effect is most evident in the over-the-air (OTA) measurements, where the receiver SNR limits the achievable EVM, even at low symbol rates. We further investigate the effect of ARoF being more sensi-

tive to non-linearities by looking at how EVM is affected by the order of QAM modulation. In Fig. 4, we present the EVM for QPSK and 16QAM, measured at point C. At 100 MBd QPSK, ARoF and SDoF achieve similar EVM. However, for 16QAM, which has higher peak-to-average power ratio than QPSK, the EVM for SDoF is lower than for ARoF, as the non-linearities of the system affects the ARoF link more severely. At 300 MBd, the effect is less pronounced as EVM for both SDoF and ARoF is limited by receiver SNR and not non-linear distortion.

Conclusion

We have compared SDoF and ARoF links that use optical frequency up-conversion to generate mm-wave carriers for wireless transmission. Through measurements, we show that below 300 MBd, SDoF is beneficial due to its increased resilience to non-linearities in the optical modulator. Above 300 MBd, decreased oversampling ratio results in higher quantization noise and consequently EVM for SDoF, making it more advantageous to use ARoF. In conclusion, the presented measurement results demonstrate the feasibility of implementing millimeter-wave D-MIMO systems using optical up-conversion together with SDoF or ARoF.

Acknowledgements

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